

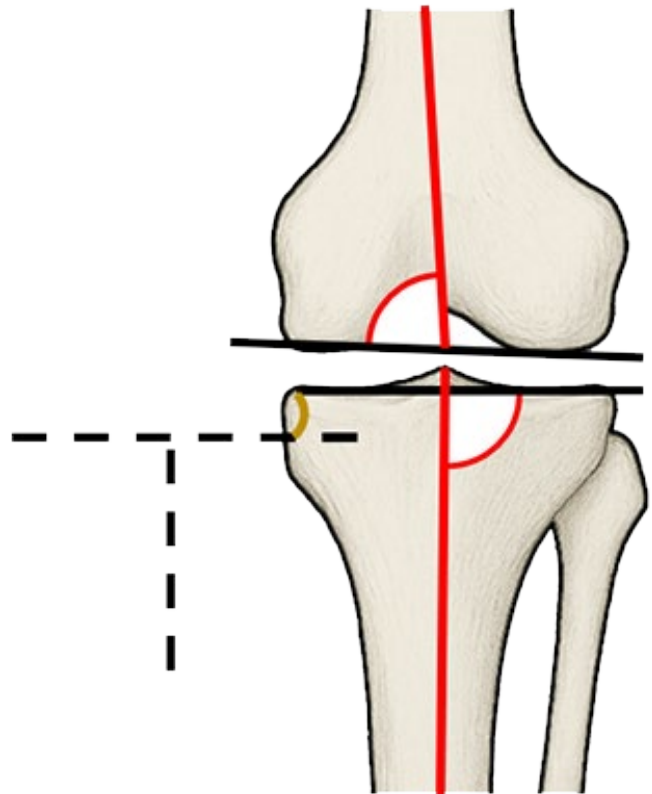
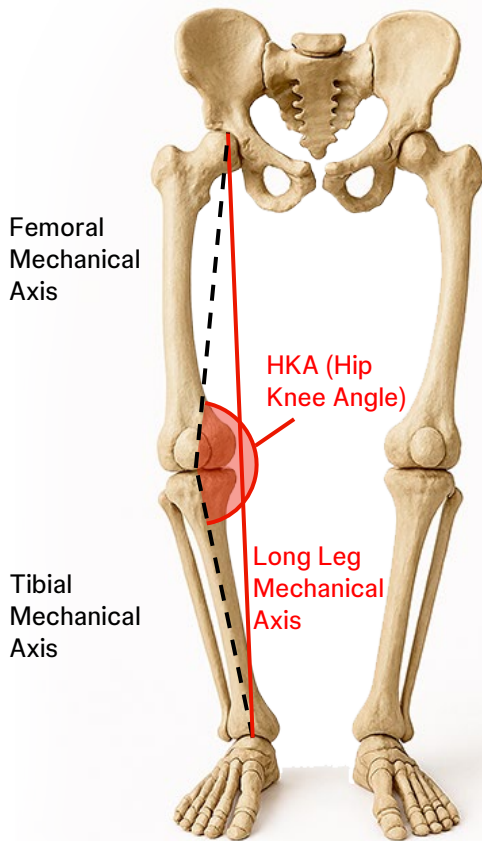


Coronal Plane Alignment of the Knee (CPAK) Classification

Optimizing Outcomes in Total Knee Arthroplasty (TKA).

Understanding native coronal alignment of the knee is critical.

The Coronal Plane Alignment of the Knee (CPAK) classification is a system designed to describe native (pre-arthritic) and arthritic knee alignment. It helps categorize different knee types based on two radiographic measurements:



The Hip-Knee-Ankle angle (HKA)

Joint Line Obliquity (JLO).

In the CPAK classification system, the arithmetic Hip-Knee-Ankle (aHKA) is used. aHKA combines the measured HKA with JLO to give a more accurate description of the knee's native alignment.

By combining the arithmetic Hip-Knee-Ankle angle (aHKA) and Joint Line Obliquity (JLO), CPAK helps surgeons identify a patient's native alignment and choose an appropriate alignment philosophy. This framework not only enhances preoperative planning but also improves intraoperative balance and postoperative function by acknowledging the wide variation in knee anatomy.

Core Concepts

In order to calculate the aHKA and the JLO, you will need the Lateral Distal Femoral Angle (LDFA) and the Medial Proximal Tibial Angle (MPTA):

Lateral Distal Femoral Angle (LDFA)

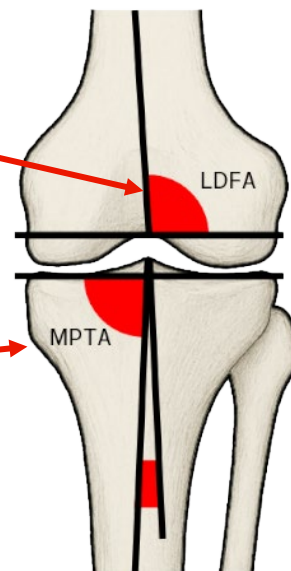
The angle between the femoral mechanical axis and the distal femoral joint line

Normal range: $\sim 85^{\circ}$ – 90°

Medial Proximal Tibial Angle (MPTA)

The angle between the tibial mechanical axis and the tibial plateau

Normal range: $\sim 85^{\circ}$ – 90°



Then, estimate the constitutional alignment using the aHKA formula, recognizing that LDFA and MPTA must be measured on long-leg radiographs

Arithmetic Hip-Knee Ankle Angle (aHKA)

$$\text{aHKA} = \text{MPTA} - \text{LDFA}$$

↑
Medial Proximal Tibial Angle

Angle between the tibial mechanical axis and the tibial plateau

↑
Lateral Distal Femoral Angle

- Negative aHKA = Varus (bow-legged)
- +** Positive aHKA = Valgus (knock-kneed)
- 0** Zero aHKA = Neutral

Joint Line Obliquity (JLO)

Affects knee balance and soft tissue tension during TKA. JLO determines the tilt of the knee joint line.

$$\text{JLO} = \text{MPTA} + \text{LDFA}$$

- D** Apex Distal ($<177^{\circ}$) = joint line tilts downward laterally
- N** Neutral (177° – 183°) = Neutral (horizontal joint line)
- P** Apex Proximal ($>183^{\circ}$) = joint line tilts upward laterally

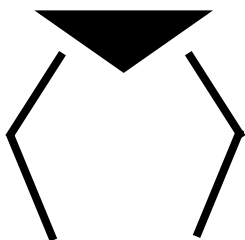
Connecting The Dots

2 Key Measurements

aHKA (limb alignment)

aHKA (Arithmetic HKA) = MPTA – LDFA

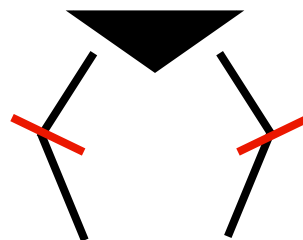
Describes limb axis: Varus/Neutral/Valgus



JLO (joint line obliquity/tilt)

JLO (Joint Line Obliquity) = MPTA + LDFA

Describes the joint line tilt: Apex Distal/Neutral/Apex Proximal



Think of aHKA as the “alignment axis” and JLO as the “tilt of the joint line”
Together they place the knee into the CPAK 9-box grid (Types I-IX)

Axis

Orientation

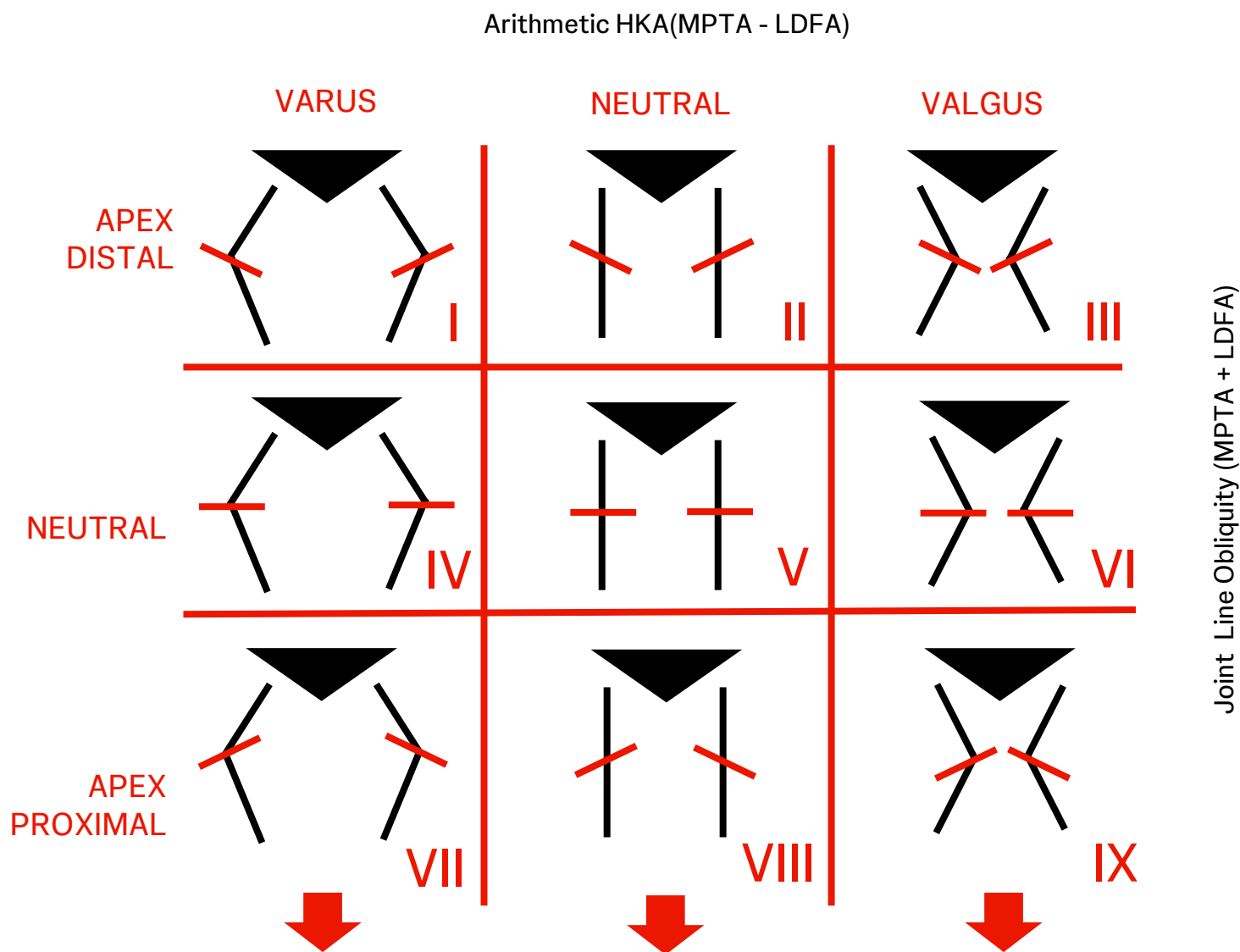
Combine to classify knee phenotype

2 Inputs – One Classification

aHKA and JLO measured separately, merged into CPAK phenotype

Nine knee classifications using a simple and reproducible method.

The 9 CPAK types from combining 3 aHKA categories (Varus, Neutral, Valgus) with 3 JLO categories (Apex Distal, Neutral, Apex Proximal):



- Type I: Varus aHKA, Apex Distal JLO

- Type IV: Varus aHKA, Neutral JLO

- Type VII: Varus aHKA, Apex Proximal JLO

- Type II: Neutral aHKA, Apex Distal JLO

- Type V: Neutral aHKA, Neutral JLO

- Type VIII: Neutral aHKA, Apex Proximal JLO

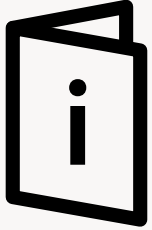
- Type III: Valgus aHKA, Apex Distal JLO

- Type VI: Valgus aHKA, Neutral JLO

- Type IX: Valgus aHKA, Apex Proximal JLO

Study Design- MacDessi et al., 2021

A comprehensive classification for coronal lower limb alignment with predictive capabilities for knee balance would be beneficial in total knee arthroplasty (TKA). This paper describes the Coronal Plane Alignment of the Knee (CPAK) classification and examines its utility in preoperative soft tissue balance prediction, comparing kinematic alignment (KA) to mechanical alignment (MA).



Background Information

- Part 1: Analysis of 1,000 knees (500 healthy, 500 arthritic) using long leg radiographs to determine CPAK distribution.
- Part 2: 138 TKA surgeries randomized to Kinematic Alignment (KA) or Mechanical Alignment (MA) to assess balance outcomes.



Key Findings

- Most common types in healthy individuals:

Type II

39%

Type I

26%

Type V

15%

- Rare types: VII, VIII, IX
- Surgical Balance at 10° Flexion:

Type I

100% KA vs
15% MA

Type II

78% KA vs
46% MA

Type IV

89% KA vs
0% MA

Study Design- MacDessi et al., 2021



Clinical Takeaways

- 1 The CPAK classification system combines arithmetic HKA (aHKA) and Joint Line Obliquity (JLO) to categorize knee phenotypes into nine distinct types, providing a standardized method to understand coronal plane alignment.
- 2 MacDessi et al. (2021) demonstrated that CPAK types I, II, and IV are the most responsive to kinematic alignment (KA), with significantly improved intraoperative balance compared to mechanical alignment (MA).
- 3 CPAK Types I and II are the most common in both healthy and arthritic populations, suggesting that many patients may benefit from alignment strategies that restore native joint lines rather than enforce neutrality.
- 4 The use of CPAK to guide surgical alignment strategies offers the potential for more personalized treatment planning, improved balance, and potentially better postoperative function and satisfaction.
- 5 JLO, as a distinct parameter from limb alignment, has independent significance in achieving soft tissue balance and functional motion—particularly in flexion.
- 6 CPAK supports the notion that 'one-size-fits-all' alignment strategies (e.g., strict MA) may not be optimal for the majority of patients undergoing TKA.

References.

MacDessi SJ, Griffiths-Jones W, Harris IA, Bellemans J, Chen DB. Coronal Plane Alignment of the Knee (CPAK) classification: a new system for describing knee phenotypes. Bone Joint J. 2021;103-B(2):329–337. doi:10.1302/0301-620X.103B2.BJJ-2020-1050.R1